

034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

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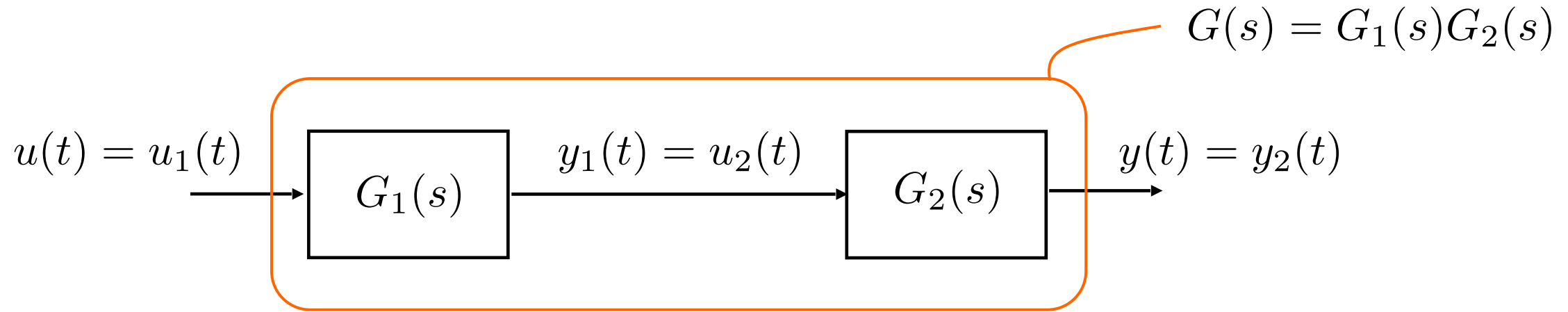
Part VI: Stability of Interconnected Systems

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Motivations:

- **Block schemes** may be useful to analyse **stability** of interconnected systems
- The **structure** and **topology** of the block scheme can be exploited under appropriate conditions
- Care should be exercised when **common factors** arise in the context of block-scheme reduction



$$G_1(s) = \frac{N_1(s)}{\varphi_1(s)} ; \quad G_2(s) = \frac{N_2(s)}{\varphi_2(s)}$$


$$G(s) = \frac{N_1(s)N_2(s)}{\varphi_1(s)\varphi_2(s)} = \frac{N(s)}{\varphi(s)}$$

- **Case 1:** no common factors among $N(s)$ and $\varphi(s)$

↳ Poles of $G(s) = \{\text{Poles of } G_1(s)\} \cup \{\text{Poles of } G_2(s)\}$

↳ $G(s)$ asymptotically stable $\longleftrightarrow G_1(s), G_2(s)$ asymptotically stable

- **Case 2:** common factors among $N(s)$ and $\varphi(s)$

- if $\text{Re poles}[\varphi(s)]$ in common < 0

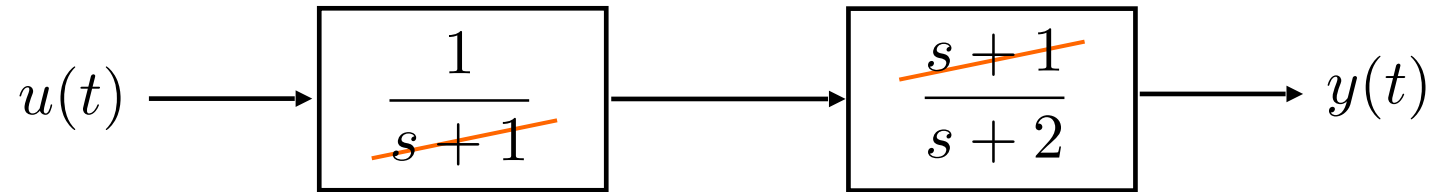
↳ **hidden internal dynamics** is asymptotically stable

- if $\text{Re poles}[\varphi(s)]$ in common ≥ 0

↳ hidden internal dynamics is not asymptotically stable

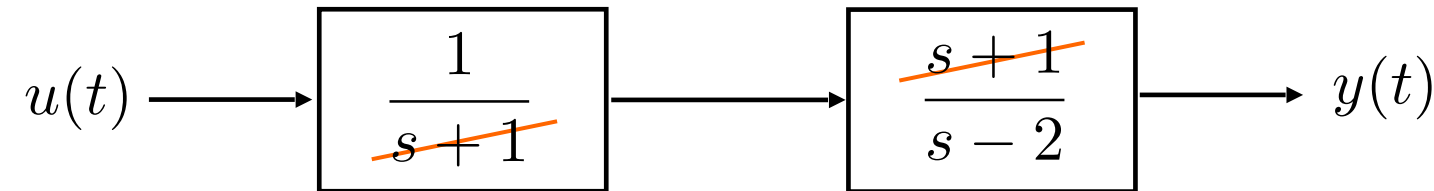
↳ stability cannot be inferred from $G(s)$

Example 1



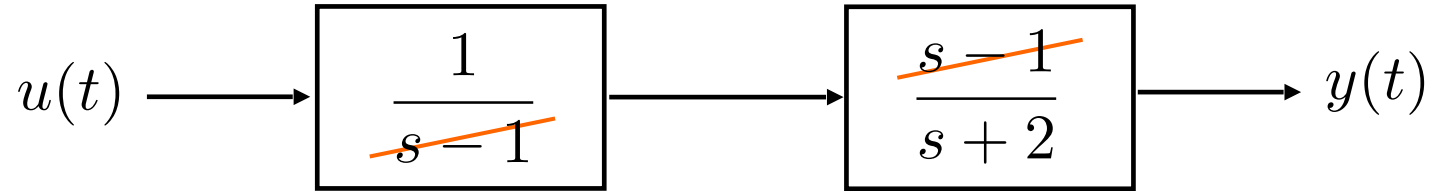
$G(s) = \frac{1}{s+2}$  system asymptotically stable and $G(s)$ shows this

Example 2



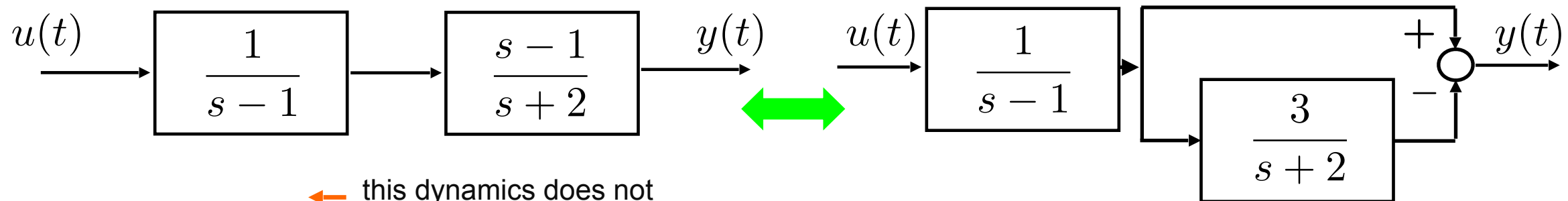
$G(s) = \frac{1}{s-2}$  system unstable and $G(s)$ shows this

Example 3



$G(s) = \frac{1}{s+2}$ ➡ system unstable but $G(s)$ does **not** show this

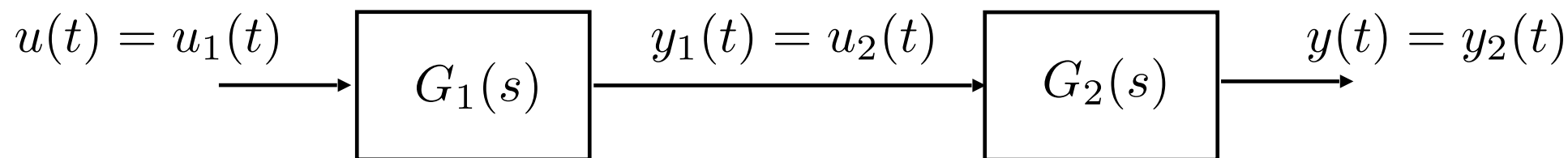
In fact:



$\begin{cases} \dot{x}_1 = x_1 + u \\ \dot{x}_2 = -2x_2 + 3x_1 \\ y = x_1 - x_2 \end{cases}$
← this dynamics does not influence the input/output relationship

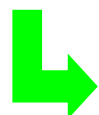
positive eigenvalue

$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \\ y = [1 \quad -1] x \end{cases}$$

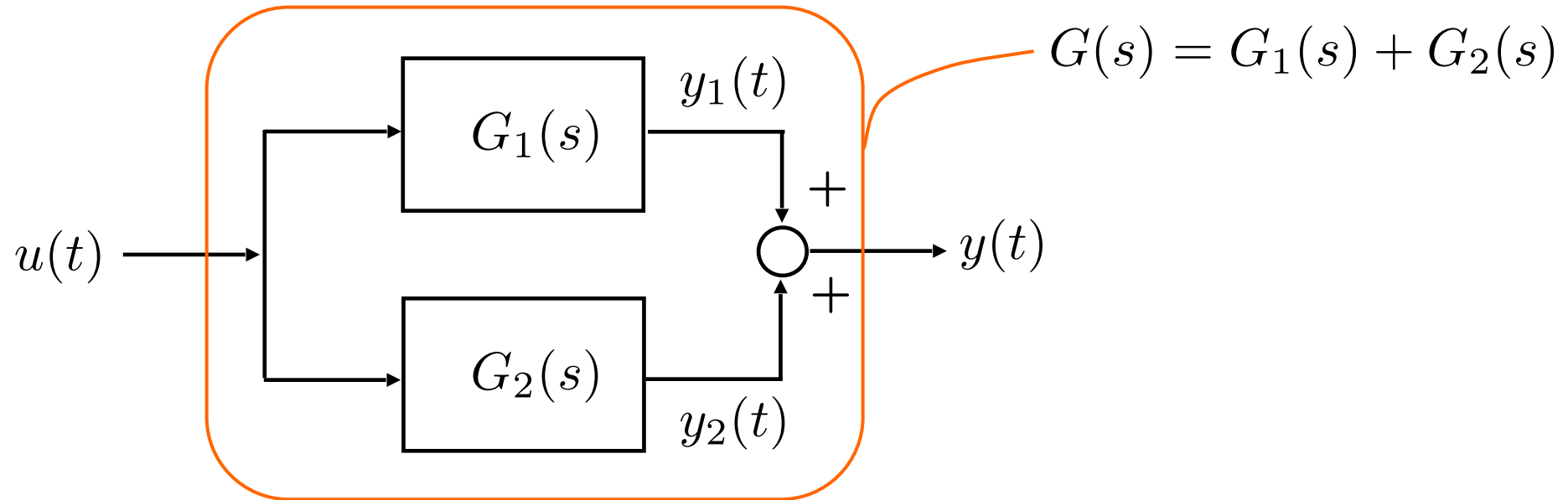


$$G_1(s) = c_1(sI - A_1)^{-1}b_1 + d_1; \quad G_2(s) = c_2(sI - A_2)^{-1}b_2 + d_2$$

$$\begin{cases} \dot{x}_1 = A_1x_1 + b_1u \\ y_1 = c_1x_1 + d_1u \\ \dot{x}_2 = A_2x_2 + b_2y_1 = A_2x_2 + b_2c_1x_1 + b_2d_1u \\ y = c_2x_2 + d_2y_1 \end{cases}$$



$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ b_2c_1 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2d_1 \end{bmatrix} u \\ y = [d_2c_1 \quad c_2] x + d_1d_2u \end{cases}$$

- Asymptotically Stable Series-Interconnected System  Asymptotically Stable Component Sub-Systems
- Systems' eigenvalues are **not modified** by series-interconnections



$$G_1(s) = \frac{N_1(s)}{\varphi_1(s)} ; \quad G_2(s) = \frac{N_2(s)}{\varphi_2(s)}$$

➡
$$G(s) = \frac{N_1(s)}{\varphi_1(s)} + \frac{N_2(s)}{\varphi_2(s)} = \frac{N_1(s)\varphi_2(s) + N_2(s)\varphi_1(s)}{\varphi_1(s)\varphi_2(s)} = \frac{N(s)}{\varphi(s)}$$

- **Case 1:** no common factors among $N(s)$ and $\varphi(s)$

↳ Poles of $G(s) = \{\text{Poles of } G_1(s)\} \cup \{\text{Poles of } G_2(s)\}$

↳ $G(s)$ asymptotically stable $\longleftrightarrow G_1(s), G_2(s)$ asymptotically stable

- **Case 2:** common factors among $N(s)$ and $\varphi(s)$

- if $\text{Re poles}[\varphi(s)]$ in common < 0

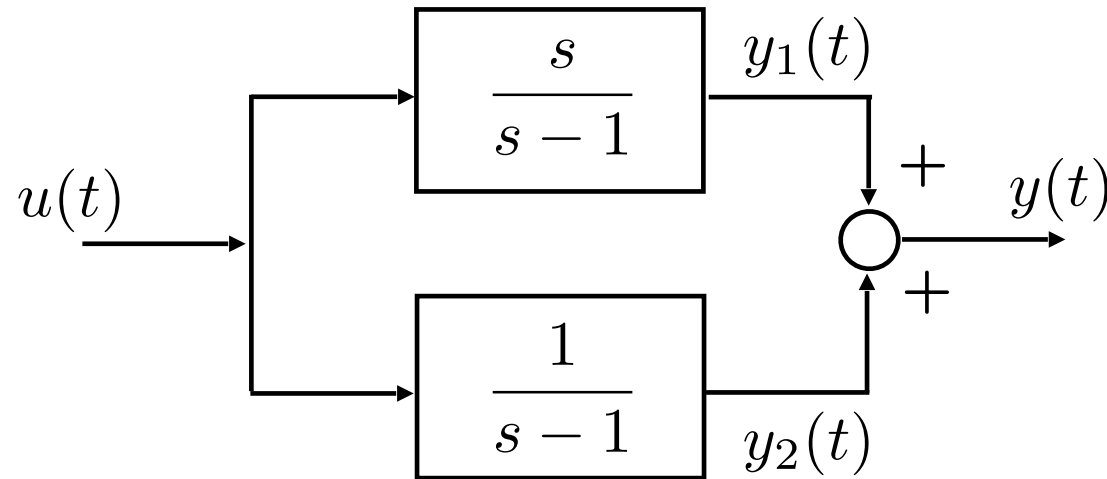
↳ **hidden internal dynamics** is asymptotically stable

- if $\text{Re poles}[\varphi(s)]$ in common ≥ 0

↳ **hidden internal dynamics** is not asymptotically stable

↳ stability cannot be inferred from $G(s)$

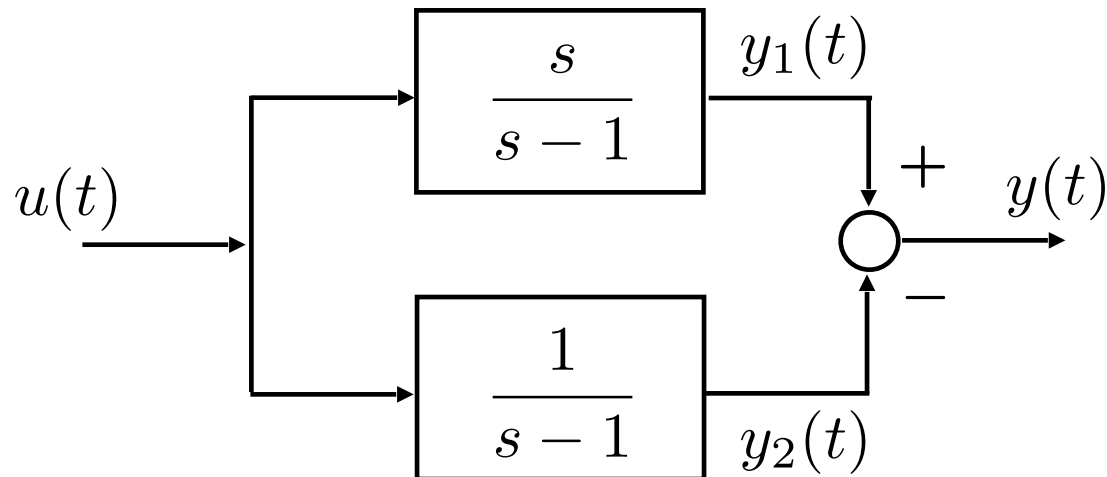
Example 1



$$G(s) = \frac{s+1}{s-1}$$

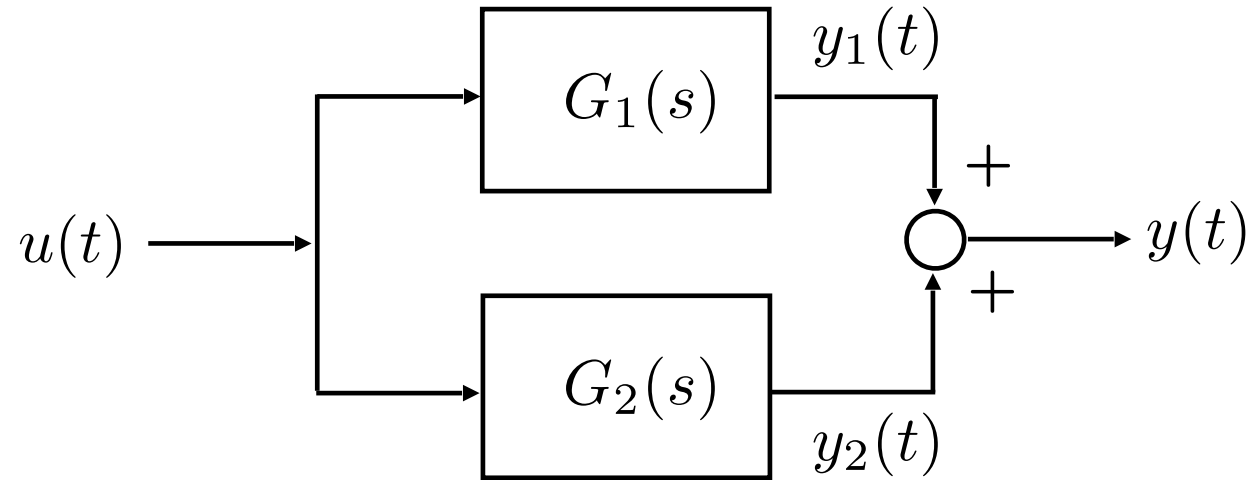
system unstable and $G(s)$ shows this

Example 2



$$G(s) = 1$$

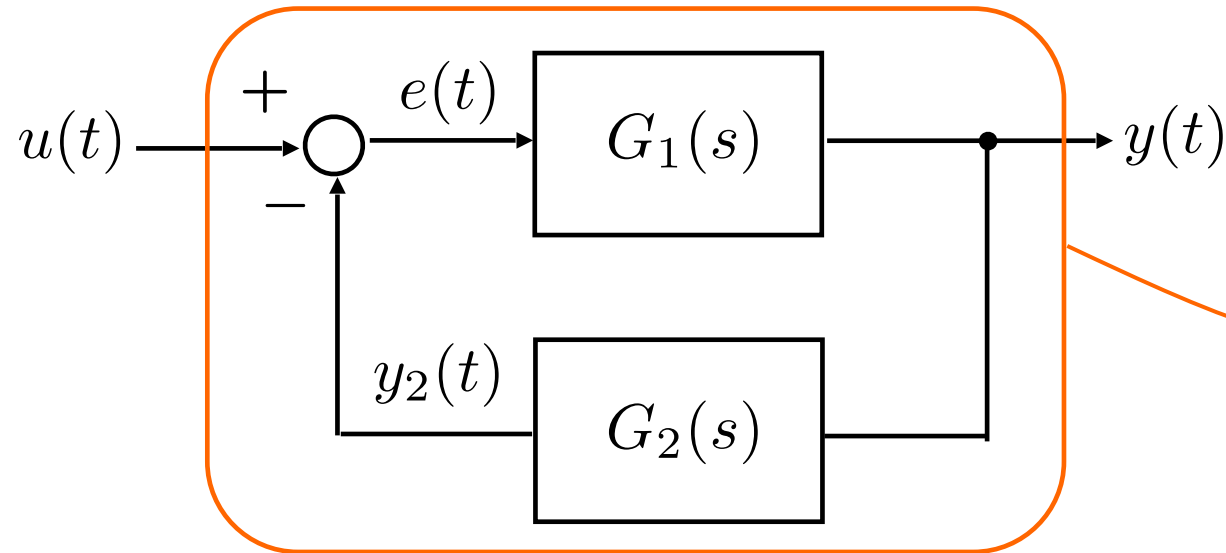
system unstable but $G(s)$ does **not** show this



$$G_1(s) = c_1(sI - A_1)^{-1}b_1 + d_1 ; \quad G_2(s) = c_2(sI - A_2)^{-1}b_2 + d_2$$

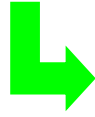
$$\left\{ \begin{array}{l} \dot{x}_1 = A_1 x_1 + b_1 u \\ y_1 = c_1 x_1 + d_1 u \end{array} \right. \quad \longrightarrow \quad \left\{ \begin{array}{l} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} u \\ y = [c_1 \quad c_2] x + (d_1 + d_2) u \end{array} \right.$$

- Asymptotically Stable Parallel-Interconnected System  Asymptotically Stable Component Sub-Systems
- Systems' eigenvalues are **not modified** by parallel-interconnections



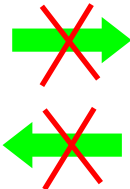
$$G(s) = \frac{G_1(s)}{1 + G_1(s)G_2(s)}$$

$$G_1(s) = \frac{N_1(s)}{\varphi_1(s)} ; \quad G_2(s) = \frac{N_2(s)}{\varphi_2(s)}$$

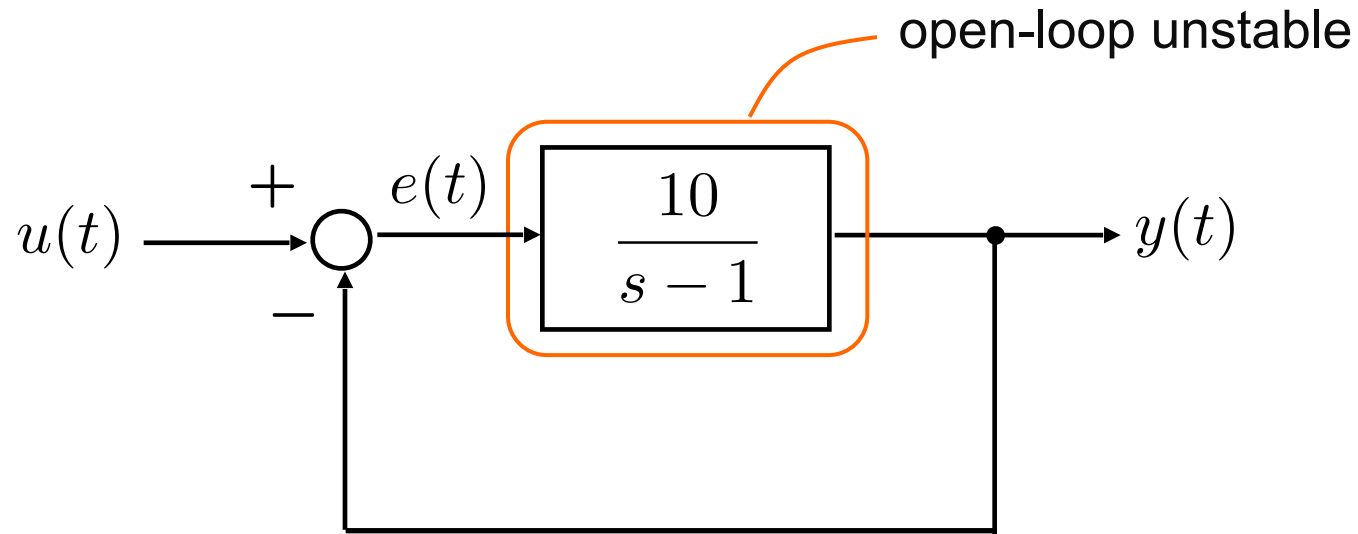

$$G(s) = \frac{\frac{N_1(s)}{\varphi_1(s)}}{1 + \frac{N_1(s)N_2(s)}{\varphi_1(s)\varphi_2(s)}} = \frac{N_1(s)\varphi_2(s)}{\varphi_1(s)\varphi_2(s) + N_1(s)N_2(s)} = \frac{N(s)}{\varphi(s)}$$

Even in the absence of common factors among $N(s)$ and $\varphi(s)$:

➤ Poles of $G(s) = \{\text{roots of } \varphi_1(s)\varphi_2(s) + N_1(s)N_2(s)\}$

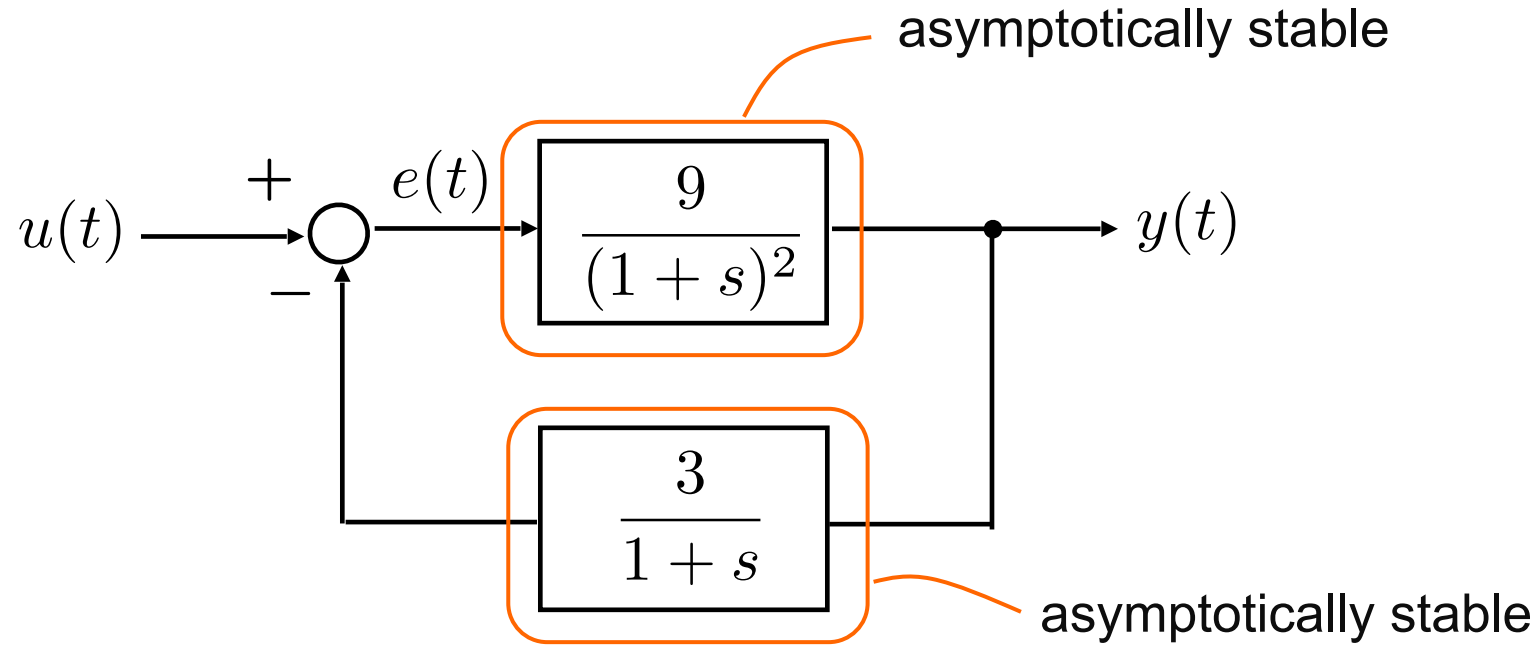
➤ $G(s)$ asymptotically stable  $G_1(s), G_2(s)$ asymptotically stable

Example 1



➡
$$G(s) = \frac{\frac{10}{s-1}}{1 + \frac{10}{s-1}} = \frac{\frac{10}{s-1}}{\frac{s-1+10}{s-1}} = \frac{10}{s+9} \quad \text{asymptotically stable}$$

Example 2



➡
$$G(s) = \frac{\frac{9}{(1+s)^2}}{1 + \frac{3}{(1+s)^2}} = \frac{9(1+s)}{(1+s)^3 + 27}$$

Hence, the characteristic polynomial is:

$$\varphi(s) = (1 + s)^3 + 27 = s^3 + 3s^2 + 3s + 28$$

The Routh table takes on the form:

1	1	3	0
2	3	28	0
3	α	0	0
4	β	0	0

r_4

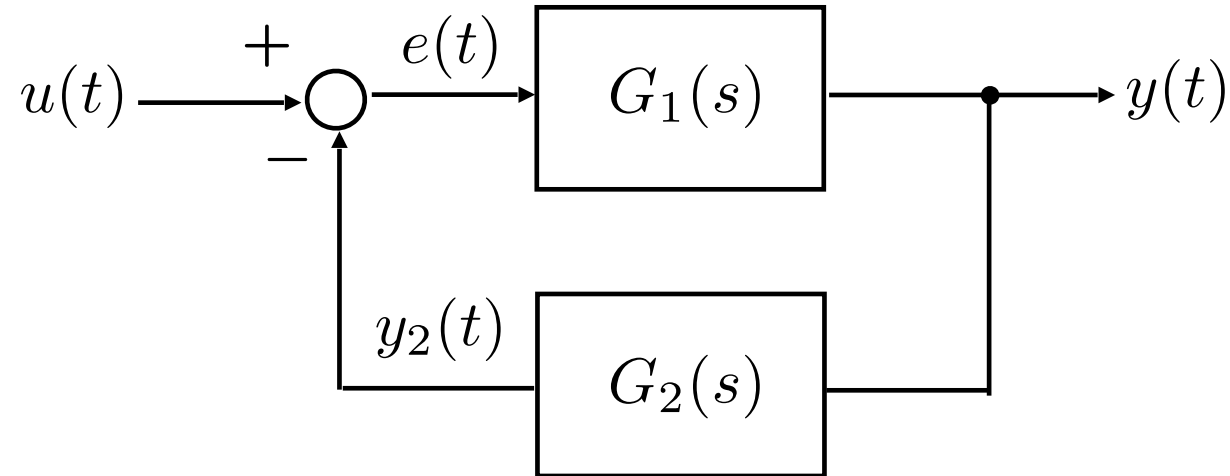
$$\alpha = -\frac{1}{3} \det \begin{bmatrix} 1 & 3 \\ 3 & 28 \end{bmatrix} = -\frac{19}{3}$$

$$\beta = 28$$



Column r_4 of the Routh table shows two sign changes and hence the closed-loop system is unstable

Assume that $G_1(s), G_2(s)$ are strictly proper:



$$G_1(s) = c_1(sI - A_1)^{-1}b_1; \quad G_2(s) = c_2(sI - A_2)^{-1}b_2$$

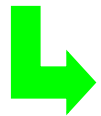
$$\begin{cases} \dot{x}_1 = A_1x_1 + b_1e \\ y = c_1x_1 \end{cases}$$

$$\begin{cases} \dot{x}_2 = A_2x_2 + b_2y \\ y_2 = c_2x_2 \end{cases}$$



$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 & -b_1c_2 \\ b_2c_1 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ 0 \end{bmatrix} u \\ y = [c_1 \quad 0] x \end{cases}$$

- Asymptotically Stable Feedback-Interconnected System  Asymptotically Stable Component Sub-Systems
- Systems' eigenvalues are **modified** by feedback-interconnections



This is key to enable design of control system that impose the desired behaviour and specifications to the controlled system